Institutions as Code: Comparative Political Institutions in a Multi-Agent Software Laboratory

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September 2026

Abstract

Comparative politics treats institutions as the rules that structure allocation, review, and accountability, yet those rules are rarely isolable in human systems. This paper presents Political Swarm, a computational laboratory in which canonical political institutions are operationalized as executable coordination protocols for large-language-model agents. Three doctrines—a stylized Westminster parliament, a federated worker-council system, and a constitutional internal market—implement the same software-engineering task on a shared engine, against two non-political controls (a managerial hierarchy and a single agent). Models, budget, task text, and a hidden eleven-check acceptance oracle are held fixed; the treatment is the decision protocol. The contribution is methodological, not confirmatory. A five-arm sweep illustrates that institutional form can be coded, logged, and compared, and that institutional bugs (agenda cycling, abandoned ownership) are themselves observable. It does not establish that any doctrine is superior, that language-model agents are political subjects, or that findings travel to human polities. Unwired essay mechanisms are disclosed rather than narrated as runtime behaviour. The repository is the replication archive.

Keywords: comparative institutions; computational social science; multi-agent systems; experimental methods; constitutional design; mechanism design.

1 Introduction

The new institutionalisms agree on a blunt claim: the rules of the game shape who may propose, who may block, who must answer, and what may not be waived (Hall and Taylor, 1996; March and Olsen, 1984; North, 1990). Empirical comparative politics has built a rich observational and experimental repertoire around that claim (Cox, 1997; Druckman et al., 2006; Lijphart, 1999; Tsebelis, 2002). What it has not usually been able to do is hold the actors, the task, and the resource constraint fixed while swapping only the constitution.

That identification problem is not a failure of political science. Human subjects bring preferences, identities, histories, and unobserved coalitions; constitutions are bundled with social structure (Greif, 2006; Pierson, 2000). Agent-based models isolate mechanisms by construction (Axelrod, 1984; Cederman,

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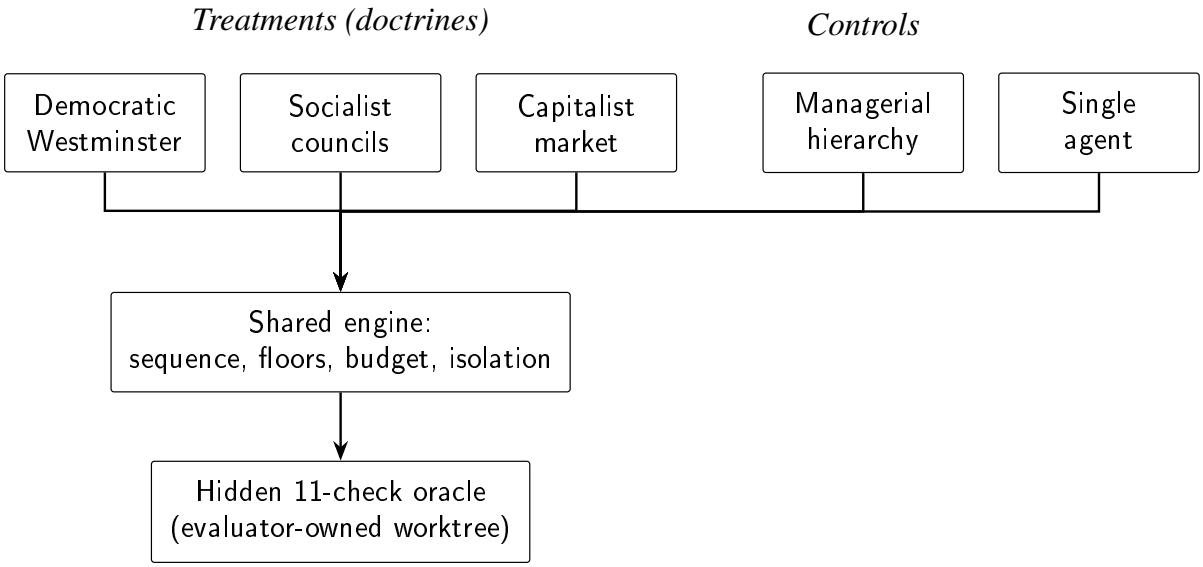


Figure 1: Identification layout. Treatments and controls share the engine, budget, task, and hidden oracle. Doctrine-specific judgement sits above the shared floors, not instead of them.

1997; Epstein, 1999), but they typically encode simple behavioural rules rather than thick parliamentary, council, or market procedure. Recent large-language-model (LLM) multi-agent systems go the other way: they give agents rich language and specialised roles, then vary models, prompts, and topology (Hong et al., 2024; Qian et al., 2024; Wu et al., 2023). Governance, when it appears at all, is a software architecture rather than a political institution.

This paper treats that gap as a methods opportunity. Political Swarm (Carr, 2026) is a laboratory in which three political doctrines and two control architectures coordinate the same coding task. The democratic arm is a constitutional parliamentary swarm in the Westminster tradition: a Speaker on admissibility, an Opposition that must be answered, a division of constituency representatives, a revising second chamber, and a gate no majority can waive (Bagehot, 1867; Dicey, 1885). The socialist arm is a federated council swarm: reconnaissance before the plan, capability-and-need allocation, recallable coordination, mutual aid, and preserved minority reports (Cole, 1920; Kropotkin, 1902; Marx, 1871; Pannekoek, 1950). The capitalist arm is a constitutional market: bids, comparative-advantage scoring, escrow, a contractual assessor with no success fee, and a regulatory floor no commercial party can waive (Coase, 1937; Hayek, 1945; Roth, 2002). Two benchmark-only controls—a managerial hierarchy and a single agent—exist so that political procedure can be compared with ordinary software organisation, not only with other political procedure.

The identifying design is simple and deliberately modest. A doctrine-neutral engine owns sequence, isolation, budgets, test integrity, and non-waivable floors. Each doctrine owns judgement: who plans, who is assigned, who reviews, and how a decision is recorded. A politically neutral orchestrator may not replace the doctrine's own procedure. A hidden oracle scores the delivered worktree independently of the swarm's self-report. Figure 1 summarises the layout.

The paper's claim is not that democracies, socialist councils, or markets "win" at software, nor that LLM agents are citizens. It is that political institutions can be written as code with enough fidelity that their characteristic evidence—divisions, minority reports, returned assignments, bids and escrow—

appears in a ledger, and that this makes institutional form a treatment in a computational experiment. A single controlled sweep (21 September 2026) is reported as a proof of concept, with the fairness defects of that sweep disclosed in the results. Repeated trials, not this $N=1$ run, are what would support comparative inference.

Comparative politics already knows how to describe constitutions. What it lacks is a setting in which those descriptions can be installed, one at a time, as the operating system of a productive task whose output can be scored without asking a human which regime they prefer. Language-model agents are a poor substitute for citizens and a serviceable substitute for boundedly rational clerks: they follow (and invent, and forget) procedures, they produce a written record, and they can be given identical instructions under different rules. That is enough for a laboratory. It is not enough for a theory of democracy.

Three design choices keep the analogy from collapsing into theatre. First, the engine is doctrine-neutral: sequence and floors do not belong to Westminster, the councils, or the market. Second, characteristic evidence is specified in advance—divisions, minority reports, returned assignments, bids—so that a run which does not produce that evidence has not instantiated the institution, whatever the folder is named. Third, two non-political controls prevent every contrast from being read as "democracy versus capitalism." If a parliament is slower than a manager, the comparison is about procedure, not about the Enlightenment.

The rest of the paper proceeds as a methods article. Section 2 situates the laboratory in comparative institutional analysis, computational social science, and multi-agent software. Section 3 maps the three ideal types and two controls. Section 4 describes the runtime. Section 5 states the research design. Section 6 reports the sweep without overclaiming it. Section 7 asks what the method offers political science. Section 8 states limitations. Section 9 concludes. Appendices separate implemented from essay-only institutions, list the oracle checks, and give replication commands.

2 Related work

2.1 Institutions as rules, and the problem of isolation

North (1990) defines institutions as the humanly devised constraints that structure interaction. Ostrom (1990, 2005) supplies the operational counterpart: an institutional statement names who may do what, under which conditions, with which sanctions, monitored by whom. The Institutional Analysis and Development (IAD) framework was built precisely so that rules-in-form could be distinguished from rules-in-use, and so that action arenas could be compared without assuming that the same noun ("council", "market", "legislature") meant the same mechanism.

Comparative research then faced a measurement problem. Lijphart (1999) arrays democracies on executives-parties and federal-unitary dimensions. Tsebelis (1995, 2002) compresses institutional difference into veto players: the number, congruence, and cohesion of actors whose agreement is required for policy change. Cox (1997) shows how electoral rules structure strategic coordination. Martin and Vanberg (2011) treats legislative procedure as a coalition-management technology. These are powerful reductions. They still observe institutions bundled with parties, voters, and history (Pierson, 2000). Laboratory experiments isolate treatments in human subjects (Druckman et al., 2006, 2011), but they cannot usually install a Westminster opposition, a recallable coordinating council, and an escrow market

as alternative constitutions for the same productive task.

Political Swarm takes Ostrom's advice literally: write the rules so that an action arena can be replayed. The cost is analogical. The agents are not citizens; the "Commons" is not a demos; prices are internal bookkeeping. The gain is that allocation, review, decision thresholds, and non-waivable floors become code paths rather than narratives about code paths.

Hall and Taylor (1996) distinguish historical, rational-choice, and sociological institutionalism. The laboratory is closest to the rational-choice branch in one respect—it treats institutions as constraints on choice—and closest to the historical branch in another: sequence matters, early refusals constrain later ones, and property in files is a form of increasing returns (Pierson, 2000). It is weakest as sociological institutionalism. There is no culture to socialise into, unless one counts system prompts, which are too thin to bear that weight. The paper therefore does not claim to test March and Olsen's logic of appropriateness (March and Olsen, 1984); it tests whether logics of aggregation, veto, and allocation can be installed as alternative operating systems for the same task.

That ambition sits beside, rather than inside, field and survey experiment. Druckman et al. (2006, 2011) document the growth of experimental political science as a way to identify causal effects of information, institutions, and elite cues on human subjects. The present design identifies the effect of *coded* institutions on *model* subjects. The internal validity target is the same; the population to which one might generalise is not. Section 7 returns to analogical inference, which is the only external-validity story this method can honestly tell.

2.2 Computational social science and agent-based models

Computational social science has long argued that digital traces and generative models let researchers study social processes that interviews and aggregate statistics cannot reach (Lazer et al., 2009; Salganik, 2018). Agent-based models are the native method for "growing" macros from micros (Epstein, 1999; Epstein and Axtell, 1996). Axelrod (1984, 1997) showed that cooperation, culture, and norms can be studied as evolving strategies. Cederman (1997) treated states as emergent rather than assumed actors.

Those models encode institutions as payoff structures, network topologies, or simple update rules. They rarely encode *procedure*: a Speaker who may not decide the substantive question; an Opposition that must name a remedy; a second chamber that revises findings but does not vote on motions; a worker's right to return an assignment; an assessor forbidden a success fee. Thick procedure is where comparative politics lives. A laboratory that wants to speak to that literature has to implement offices, not only incentive parameters.

The distinction is worth labouring. Axelrod's tournaments vary strategies inside a fixed game (Axelrod, 1984). Cederman varies agent types inside a geopolitical space (Cederman, 1997). Epstein asks whether a macro regularity can be grown from local rules (Epstein, 1999). Political Swarm varies the game itself—the offices, the agenda, the property rule, the price—while holding the "players" as constant as a shared model roster allows. In IAD terms, it treats the action arena as the treatment and the participants as (noisy) controls (Ostrom, 2005). That is closer to constitutional political economy (Buchanan and Tullock, 1962) than to evolutionary game theory, even though the implementation happens to use language models.

2.3 Mechanism design and markets as engineered institutions

A parallel tradition treats institutions as engineered allocation devices. Roth (2002) describes the economist as engineer: theory, experiment, and computation in the service of rules that have to work when fielded. Hayek (1945) is the knowledge argument for prices; Coase (1937, 1960) is the argument that firms exist when internal coordination is cheaper than repeated contracting, and that externalities follow transaction costs. Olson (1965) and Buchanan and Tullock (1962) add the public-goods and constitutional-calculus problems: shared infrastructure is under-provided, and the rules for choosing rules are themselves a design object.

The capitalist arm of Political Swarm is an explicit attempt to make that tradition runnable inside a coding swarm: an exchange, capital accounts, contract shapes, insurance, and a regulator confined to rights, contract enforcement, and a safety floor. It is not a model of actually existing capitalism. It is a constitutional market in Roth's sense—a designed game whose pieces can be inspected.

2.4 Multi-agent systems, without governance as the treatment

Classical multi-agent systems already distinguished protocols from agents (Wooldridge, 2009). The last three years of LLM swarms have revived that architecture with linguistic agents. ChatDev organises a virtual software company through a chat chain (Qian et al., 2024). MetaGPT encodes standard operating procedures as meta-programming (Hong et al., 2024). AutoGen supplies a general conversation framework (Wu et al., 2023). Multi-agent debate can improve factuality (Du et al., 2024). In all of these systems the scientific variable is almost always the model, the prompt, the role roster, or the communication topology.

That is a reasonable engineering research programme. It is not a comparative-politics research programme. If every swarm is a firm with a CEO, or a pipeline with stages, then governance is held constant while models vary. Political Swarm inverts the usual experiment: models, task, and budget are held as constant as the stack allows, and the decision system is the treatment. The two control arms exist because a Westminster parliament should be compared not only with councils and markets but with the organisational forms software engineering already uses.

Debate swarms (Du et al., 2024) are the nearest neighbour and still not the same object. They vary whether agents argue; they do not vary whether argument is an Opposition with a duty to answer, a council minority report, or a priced dispute before an assessor. The institutional content of disagreement is the variable here. Disagreement as a prompting trick is not.

3 Ideal types and their mapping

The three doctrines are stylized types, not empirical descriptions of the United Kingdom, the Paris Commune, or any existing market society. Each type is specified in a design essay and then implemented, only in part, as a `SwarmStrategy`. Table 1 summarises the intended mapping along five functions that comparative institutionalists already use: organisation, allocation, review, decisions, and accountability. Appendix A records which of those functions are actually wired.

Table 1: Three political doctrines as software institutions. These are mechanisms under investigation, not claims about real-world regimes.

Dimension	Democratic	Socialist	Capitalist
Organisation	Two chambers, cabinet, shadow cabinet, neutral Speaker	Worker councils federated by specialist guilds	Internal task exchange; firms subcontract when cheaper than contracting
Allocation	Committee planning; seats by election, appointment, and lot	Work by capability, resources by assessed need; right of return	Competitive bids on comparative advantage
Review	Opposition brief, government reply, constituency division	Peer review outside authorship; overruled objections preserved	Assessor with no success fee; court for unsettled disputes
Decisions	Risk-scaled thresholds above a non-waivable constitutional gate	Subsidiarity, delegate recall, evidence gate	Priced scarcity, contract shape, regulatory floor
Accountability	Confidence, recall, Question Time, report against the mandate	Recall, independent diagnosis, commons report	Profit and loss, insolvency, market report
Characteristic evidence	Divisions, minority reports, Speaker rulings	Reconnaissance, returned assignments, mutual aid, dissent	Bids and awards, priced consumption, levies, refused rescues

3.1 Democratic: mixed government as a coding constitution

The democratic design does not treat democracy as universal voting on every edit. It follows the mixed-government reading of Westminster: legislature, executive, organised opposition, committees, a neutral presiding officer, and a professional continuity layer, under a constitution that binds the majority (Bagehot, 1867; Dicey, 1885; Madison, 1788; Montesquieu, 1748). Offices are constituted from the run identifier so that election, appointment, sortition, rotation, and recall are re-derivable from the ledger. The Speaker screens motions for admissibility and amends overlapping mandates so that each file has one owner; the office is barred from striking the last motion on the paper, because a procedural veto of the entire programme would be a substantive decision. The Opposition must name a remedy that reaches the next attempt; a standing objection after the House has decided becomes a minority report rather than a hidden veto. The House of Review sits on findings, not on motions. Acceptance is a deterministic constitutional gate, independent of ordinary confidence.

Two further facts matter for interpretation. First, the Commons division is a formula, not a deliberative sitting: there is no modelled debate on the floor. Second, several named Westminster pieces—a two-chamber vote on constitutional amendment, a Security Committee, a working prime minister, Question Time, and an alternative programme—are nominal or routed to the human (Appendix A). The arm is a *partial* operationalization of the essay, which is itself a partial operationalization of a historical constitution.

Partial is not the same as fake. Sortition for ordinary offices and election for competence-sensitive ones is an Aristotelian distinction the code actually draws. Organised opposition with a duty to answer is a Bagehot distinction the code actually draws. A gate independent of confidence is a Dicey distinction the code actually draws. What the arm does not draw is a talking Commons. Readers who need floor debate to call a system a parliament should treat this as a committee-centred Westminster, closer to Martin and

Vanberg (2011) than to a C-SPAN feed.

3.2 Socialist: federation without a managerial class

The socialist design starts from a question shared across otherwise opposed traditions: how specialised workers coordinate a common project without letting coordination harden into a separate class (Cole, 1920; Kropotkin, 1902; Marx, 1871; Pannekoek, 1950). The recommended answer is a federated council swarm. Worker councils inspect the repository before the plan is written. Specialist guilds advise across councils. A coordinating council is drawn by lot from the run identifier, bounded by mandate, and in principle recallable. Allocation follows assessed need and subsidiarity; a worker may return an assignment it cannot take; mutual aid is dispatched only on an actual block. Dissent that loses is kept. The commons report is filed at the end of finding disposition so that it is built into the delivery packet.

Again the implementation is thinner than the essay. Recall grounds beyond misallocation are not enforced; the deciding coordinator is a single persona; planner recall takes effect on the next run, not this one. What *is* enforced is the production-side core: reconnaissance, need-weighted assignment, the right of return, mutual aid, minority reports, and a constitutional gate that will not let correctness be voted into existence.

The right of return deserves emphasis because it is easy to misread as inefficiency. In a managerial control, a blocked worker is a failed assignment. In the council doctrine it is a constitutional event: the worker may refuse work it cannot take, mutual aid may be dispatched, and an independent diagnosis precedes reassignment after repeated failure. That is closer to Kropotkin (1902) than to a retry loop, even when the implementation is a retry loop with extra records. The records are the institution.

3.3 Capitalist: a constitutional market, not a free-for-all

The capitalist design is Hayekian about knowledge and Coasean about the firm, with a Nozick-like regulator that is supposed to enforce rights rather than plan production (Coase, 1937; Friedman, 1962; Hayek, 1945; Schumpeter, 1942; Smith, 1776). Agents bid; awards favour comparative advantage rather than the strongest absolute bidder; scarce compute is priced; tightly coupled work may be subcontracted inside a temporary firm; an assessor is paid without a success fee; escrow releases under a regulatory floor.

The market is constitutional because three commercial parties in agreement still cannot waive safety, test integrity, or acceptance evidence. It is also incomplete. Public-goods and venture funds accumulate levies but issue no grants; credit drawn to cover an insurance-premium shortfall is never repaid; warranty periods are priced but not counted down; bid-collusion detection is a logged flag, not a block. Those gaps are design facts, not footnotes to be silently closed in the results narrative.

They are also theoretically informative. Olson (1965) predicts that public goods inside a market will be under-provided unless someone is assigned to supply them. A treasury that levies but never grants is an unusually pure illustration: the accounting identity exists, the disbursement function does not. Coase (1937) predicts that firms appear when the cost of using the exchange exceeds the cost of internal direction; the capitalist arm therefore permits subcontracting rather than forcing every edit through the auction. Self-dealing is blocked in a single run's register—a firm may not assess the contract it was paid for—because

that conflict is observable without cross-run history. Collusion among nominally independent bidders is not, and the runtime refuses to pretend otherwise. Honesty about what a constitution can see is part of the constitutional design.

3.4 Controls: hierarchy and the individual

The managerial control is a conventional software organisation: a roster, capability scores, a planner, implementers, and a reviewer, without parliament, councils, or exchange. The single-agent control collapses planning, implementation, and self-review into one persona. Neither is a political doctrine, and neither is offered through the user-facing start command. They exist so that any cost of political procedure can be compared with the organisational forms the rest of the literature already uses. If a Westminster swarm is slower than a manager, that is a result about procedure, not a result about "democracy versus capitalism."

4 The laboratory

4.1 Engine versus doctrine

The runtime makes a constitutional distinction that political theory would recognise as the difference between the rules of order and the sitting government. The engine owns *sequence and floors*. It creates an isolated Git worktree, records a baseline, asks the doctrine to plan and declare ownership, then loops over assignment, execution, checks, review, decision, and recording. Concurrency is allowed only across tasks with disjoint declared ownership. After the loop, integration, independent verification, finding disposition, and doctrine-level acceptance are serial barriers. The engine never substitutes its judgement for the doctrine's procedure. That boundary is identical in the three philosophy documents and is traced, mechanism by mechanism, in the orchestrator contract (Carr, 2026).

Each doctrine implements the same barrier ports (integrate, verify, dispose, accept, remediate) with its own offices (Table 2). Who convenes is political; that the barriers happen, and in that order, is not.

4.2 Rules-in-form, rules-in-use, and deterministic offices

Ostrom (2005) insists that rules-in-form and rules-in-use can diverge. The laboratory makes that divergence inspectable. Some offices are deterministic functions of the run record: elections and sortition seeded from the run identifier, Commons divisions computed from a formula, constitutional gates that do not call a model, assignment returns that ride an implementer dispatch at no extra cost. Running those offices through an LLM would make them unauditable without making them better. Other offices weigh evidence and therefore dispatch a model: select committees, Opposition briefs, government replies, the House of Review, council reconnaissance, mutual aid, contractual assessment. The pattern implied by the shared orchestrator contract is simple. A body that applies a rule does not get a model. A body that judges evidence does.

Two placements are forced by the engine rather than chosen for local colour. The House of Review sits at finding disposition, not at acceptance, because a second chamber expressing itself through `accept()` would hold a veto the doctrine withholds from it. The socialist commons report is written at the end

Table 2: The same acceptance barriers, three institutions. The engine requires the ports; the doctrine fills them.

Barrier	Democratic	Socialist	Capitalist
Integrate	Integration Committee, answerable to the Commons	Integration council of producers	Separately procured integration assessor
Verify	Verifier appointed by the Commons who did not author	Independent peer outside the authorship chain	Contractual assessor with no success fee
Dispose findings	Majority report; minority reports preserved	Common ledger; dissent preserved	Acceptance schedule with priced obligations
Accept	Constitutional verification gate, then assent	Recorded acceptance on reproducible evidence	Escrow release under the regulatory floor
Remediate	Motions recommitted for scrutiny	Returned to the responsible council	Warranty claim; repeat failure relists

of disposition rather than in `accept()`, because the delivery assembler reads doctrine notes before acceptance runs; a report filed later would disclose nothing while looking as though it had. Forced placements are the laboratory equivalent of constitutional conventions (Dicey, 1885): they are not in the essay's organisational chart, but violating them would silently change the constitution.

Run state is similarly constitutional. A swarm may be running, parked on a budget or human decision, awaiting independent acceptance, completed, in remediation, interrupted, terminated incomplete, failed, or stopped. Partial work may not be presented as accepted. Resource exhaustion parks the run; it does not, by itself, end it. Those distinctions are the difference between a laboratory and a demo.

4.3 Floors as coded vetoes

Tsebelis (2002) treats a constitution as a special veto player: it raises the threshold for change. Political Swarm encodes two such constraints above every doctrine. First, no strategy may delete, skip, or weaken an existing test to obtain a pass; the engine inspects the staged diff before the check runs and opens a human `test_waiver`. Approving the waiver does not land the weakening: it only requeues the task, and the guard runs again. Second, findings on named grounds—security, safety, privacy, licensing, human control, test integrity, acceptance evidence—are marked non-waivable. An ordinary majority, a council vote, or a paid contract cannot disposition them as accepted without a remedy. Socialist and capitalist strategies call the same marker as the democratic House of Review. Three constitutions, two prohibitions, one implementation, so the floors cannot drift apart by doctrine.

This is a stricter analogue of constitutional review than most parliamentary models use. In Tsebelis (2002), a constitutional court is a veto player whose congruence with the legislative majority varies. Here the floor is perfectly incongruent with any doctrine that would like to ship: it is not an agent with preferences, it is a pattern match plus a prohibition. That is both a strength and a weakness. The strength is auditability. The weakness is the false positive already noted, and the deeper weakness that a constitution which cannot learn is closer to a tripwire than to a court. The laboratory currently offers tripwires. Building a court—an office that interprets the grounds—would be a different, and more sociological,

experiment.

Minority reports are the other floor. A disposition that drops a marked dissent is restored. Overruled objections remain in the delivery packet. That is an Ostrom-like monitoring rule: the record of disagreement is part of the common-pool information, not an embarrassment to be garbage-collected.

4.4 Packets, isolation, and the human remainder

Between model instances the engine passes a synchronisation packet: base revision, decisions, ownership map, changed-file manifest, evidence, open findings. At the end of a run the doctrine submits a delivery packet: evidence, exact check commands and results, findings with dispositions, minority reports, limitations, residual risks. The run reaches `awaiting_acceptance`, never `completed`, on the swarm's own say-so. An independent check traces every acceptance criterion to evidence, reads the complete diff, and either completes the run or returns concrete defects to the *same* swarm. Writes into the user's repository require a further, explicit confirmation.

This is a laboratory with a sovereign. In live use the sovereign is the human. In the headless benchmark the user's selected model stands in for doctrine escalations, seeing the task, the checks, and the diff, and naming no doctrine; budget and wall-clock extensions are a fixed per-arm allowance. Either way, political procedure does not get the last word on merging code. That is a feature of the identification strategy, not an admission that the doctrines failed to be political.

Worktree isolation is a Git boundary, not an operating-system sandbox. Generated commands are untrusted. The Model Context Protocol cannot cryptographically distinguish a human request from a model's request to apply. Those are ethical and safety facts, returned to in Section 8.

5 Research design

5.1 Task

Every arm receives the same starting repository and the same goal: a small Python command line program, `alloydamage`, that computes the neutron displacement-damage cross-section of an Fe–Cr–Ni alloy from IAEA-traceable elemental data,

$$\sigma_{d,\mathrm{alloy}}(E) = \sum_i x_i \sigma_{d,i}(E),$$

with documented conversion from weight percent to atomic fraction, a common interpolated energy grid, no silent extrapolation, CSV and plot outputs, and provenance reachable from program output. The task is bounded and non-trivial. It requires numerical care, refusal behaviour, and packaging. It is not a political task. That is the point: the politics is in the coordination, not in the subject matter.

An engineering task is used for three methodological reasons. First, it yields an independent, quantitative outcome—the oracle—that does not require a human coder to prefer one doctrine's style. Second, software work has natural analogues to the functions in Table 1: plans are agendas, files are property, tests are monitoring, merges are collective decisions. Third, the same task can be given to a managerial hierarchy and a single agent without changing its meaning, which a "simulate a parliament" task could not. The cost is that readers may mistake the paper for a software-engineering bake-off.

The oracle is a measurement instrument, not the explanandum. The explanandum is whether and how institutional form changes the path to that measurement.

5.2 Treatments, controls, and what is held fixed

The five strategies share the engine, the OpenRouter roster, the per-run budget, and the task text. Every dispatcher role uses `openai/gpt-oss-20b` with `deepseek/deepseek-v4-flash` as fallback, except the implementer, which uses `z-ai/glm-5.2` with the same fallback. The sweep reported below used an eight dollar budget and a 150-minute wall-clock cap, run sequentially to avoid provider contention. Strategy order can be rotated across trials; this paper reports one sequential sweep.

What is *not* held fixed, and should not be, is the number of institutional model calls. A Speaker ruling, an Opposition brief, council reconnaissance, or a bid round is the treatment. Equalising call counts would erase the institutions.

The sovereign is the remaining identification threat. Doctrines escalate at different rates: a Westminster arm can open a vote-threshold or confidence decision; a market arm can open a spend approval; both can hit the engine's final-diff and test-waiver ports. In the benchmark, those escalations go to the user's selected model with no doctrine name and no tools, and the rationale is kept in the evaluator ledger rather than being fed to the next implementer. Arms that escalate less therefore do not receive a stronger model's engineering advice as a side payment. Sovereign cost is reported per arm because it is harness spend, not doctrine spend, and because a constitution that asks the sovereign more often is a different constitution. Budget and time extensions are a fixed per-arm allowance, not a first-come pool. Those details are tedious, and they are the difference between a treatment and a confound.

5.3 The hidden oracle and blind scoring

Per-task checks ask a narrow regression question against the latest accepted tree. Final required checks are strict. Separately, the harness runs an evaluator-owned oracle of eleven checks against the live worktree before doctrine acceptance (Appendix B). The swarm never sees the oracle's code or reference data before that barrier. Failed checks return as findings with origin `oracle`, equally for every arm. "Tests (oracle)" in the results is this independent count, not the swarm's self-report.

A two-pass rubric further separates engineering quality from governance process. Pass 1 scores blinded candidate directories on numerical correctness, nuclear-data handling, tests, documentation, simplicity, plots, and provenance, plus a small resource-efficiency weight, without strategy names. Pass 2 opens the ledger. The engineering scores are not to be revised after unblinding. That is a standard experimental hygiene borrowed from blinded coding studies and from experimental political science's insistence that the outcome measure not be the treatment (Druckman et al., 2011).

5.4 Prompt neutrality as a design-based check

A recurring threat in LLM experiments is that the prompt, not the institution, carries the answer. Earlier Political Swarm sweeps gave the democratic arm task-specific headers, provenance tokens, and oracle-mirroring checks that the other arms did not receive; shared layers also leaked alloy-specific text. Those runs are not comparable and are not reported as evidence here.

Table 3: One controlled sweep, identical task and roster. Oracle scores are independent of self-report. The italic row is not part of the sweep.

Governance	Oracle	Cost (USD)	Time	Incomplete	Defects
Democratic	1/11	1.56	2h10m	1	10
Socialist	10/11	0.48	33m	1	1
Capitalist	10/11	1.47	1h25m	1	1
Managerial	11/11	0.52	29m	0	0
Single-agent	10/11	0.54	35m	1	1
Democratic re-run	*11/11*	*0.66*	*1h03m*	*1*	*0*

A subsequent review found remaining inequalities: the capitalist implementer was told to write Node.js on a Python task; democratic and single-agent implementers did not see the whole goal; implementer temperatures differed; sovereign rejection rationales coached only the arms that escalate; the extension allowance was a first-come pool; democratic and socialist arms could read memory left by earlier runs. Those defects were removed. A unit test now fails if any strategy or the dispatcher mentions the benchmark vocabulary, if an implementer prompt lacks the goal or the shared reminders, if a prompt names a language or test framework, or if an arm samples its implementer differently. Knowledge is scoped per run.

The 21 September 2026 sweep still contained several of those inequalities. It is reported below with that caveat, as a laboratory demonstration, not as a clean estimate of institutional treatment effects.

5.5 What would constitute a confirmatory design

A confirmatory paper would pre-register the five-arm contrast, fix the code at a tagged commit, run K sequential trials with rotated order, report oracle passes, cost, duration, completion status, and process traces, and analyse ledgers as event data. This paper is not that paper. It specifies the laboratory so that that paper can be run.

The smallest honest K is larger than one and smaller than a public-opinion survey. Language-model output is overdispersed; institutional call sequences are path dependent; provider congestion is a time-varying confounder. Five to ten sequential trials on a frozen commit, with rotated order and the neutrality tests green, would support descriptive contrasts (medians, ranges, characteristic failure modes). Dozens would still not support a claim about real parliaments. The method's comparative advantage is mechanism isolation, not sample size in the survey sense.

6 Results: a proof-of-concept sweep

Table 3 reports the 21 September 2026 sweep (commit `1ca47a7`), plus one democratic re-run after a procedural bug was fixed. Human intervention is coded 1 when the run ended `terminated_incomplete` rather than `completed`—that is, when bounded remediation stopped short of delivery—and 0 otherwise. Defects are 11 minus oracle passes.

Three observations are licensed; a fourth is not.

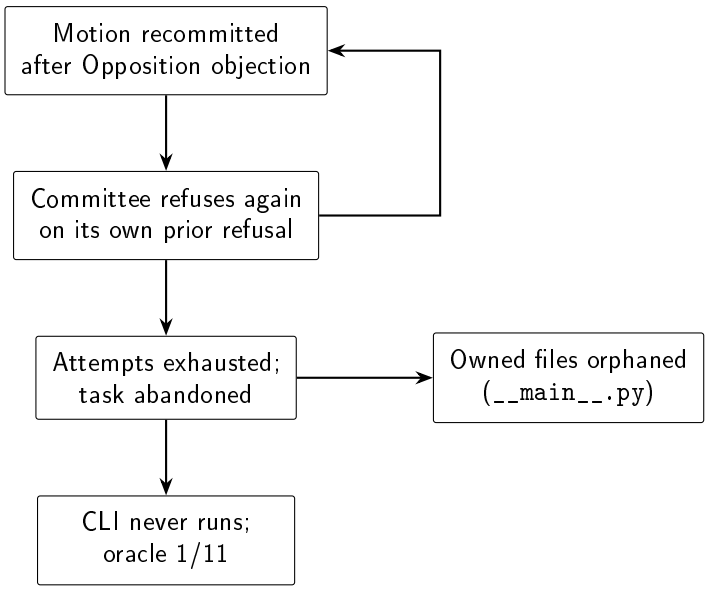


Figure 2: Democratic stale-refusal loop on the 21 September 2026 sweep. A committee treated its own prior refusal as sufficient grounds to refuse again. Agenda cycling plus exclusive file ownership produced an orphaned entry point and a collapsed oracle score. The loop is an institutional failure mode, not evidence about democracies.

First, the laboratory runs. All five arms produce ledgers, ownership maps, cost traces, and a worktree that can be oracle-scored. That is the methods result.

Second, institutional form is visible in process, not only in the score. The socialist arm's characteristic evidence is reconnaissance, returned assignments, and preserved dissent; it missed the plot check. The capitalist arm's evidence is bids, awards, and priced consumption; it missed provenance. The single agent missed provenance as well. Those single misses are exactly the details that earlier, non-neutral prompts had spelled out—a reminder that coaching had been doing institutional work.

Third, the managerial control completed 11/11 without stopping, cheaper and faster than the political arms in this sample. On $N{=}1$ that is not a finding that hierarchy dominates democracy. It is a reminder of why the controls exist. Political procedure has a call tax. Whether that tax buys review quality is an empirical question for repeated trials, not a conclusion from one afternoon.

The fourth, unlicensed, observation would be that "socialism beat democracy." It did not, in any sense this design can support. The democratic 1/11 traces to a generic procedural defect, not to missing nuclear knowledge. A recommitted committee kept refusing a motion on the strength of its own earlier refusal, while that motion's checks passed and the Opposition approved, until attempts ran out (Figure 2). The abandoned motion owned the program entry point. At the time, the engine could not hand an abandoned task's files to another task, so the module never ran. Four other motions were approved and their tests passed. After the stale-refusal loop was fixed, a re-run scored 11/11 in 63 minutes at $0.66—and still ended incomplete, because the doctrine abandoned a nuclear-data motion whose unit tests never passed, even though the delivered program covered what that motion was for.

That sequence is the most politically interesting result in the table, and it is not a score. It is what Pierson (2000) would recognise as a small increasing-returns trap: an early refusal becomes the reason for the next refusal. It is what Tsebelis (2002) would recognise as a veto player—here, a committee—

whose agreement could not be obtained even after the factual predicate had changed. It is also what the ownership rule made catastrophic: exclusive property in files, without succession on abandonment, turned a failed motion into a missing program. Subsequent engine work transfers orphaned paths to a live task. The political reading is that property rules and veto rules interact. A methods laboratory that could not produce that sentence from a ledger would not be worth running.

6.1 Process traces by arm

Scores compress. Ledgers do not. On this sweep the socialist arm spent reconnaissance tokens before the plan existed, which is the point of a council that inspects production rather than receiving a plan from above. Its single oracle miss was the plot. The capitalist arm allocated by bid and paid an assessor; its miss was provenance, a public good in Olson's sense that a market without working grants is liable to under-supply. The single agent, which has no market and no council, missed provenance too, so the under-supply is not unique to prices; it is unique to no one owning the public record. The managerial arm assigned from a roster, reviewed, and finished. It is the cheapest complete delivery in the table, and the only one that did not stop incomplete.

None of those traces is a regime ranking. They are existence proofs that the characteristic evidence named in Table 1 can appear, and that when it fails to appear—when a Commons does not deliberate, when grants do not disburse—the paper can say so rather than inferring the missing institution from the name of the folder.

7 What the method offers political science

7.1 Isolable institutions

The first offering is identification in the IAD sense (Ostrom, 2005). An action arena—a coding task with a budget—can be replayed under alternative rule configurations. The actors' "preferences" are noisy and not theoretically clean, but the rules are inspectable. Comparative statics that are usually cross-national (more veto players, more delay; more opposition rights, more disclosed risk; more price signals, different allocation) become within-laboratory contrasts.

Because the rules are code, they can also be ablated. One can disable the Opposition's duty to name a remedy, or the worker's right of return, or the assessor's ban on success fees, and ask which piece of the type was load-bearing. That is closer to the way Ostrom (1990) catalogues design principles than to a three-chimaera horse race. The present paper does not run those ablations. It argues that they are now well-defined experiments rather than thought experiments.

A related offering is measurement. Institutionalists often infer procedure from outcomes, or from texts that may not be rules-in-use. Here the ledger is the rules-in-use. If the Commons does not deliberate, the paper can say so. If grants do not disburse, the paper can say so. The gap between essay and runtime is not a scandal; it is data about how much of a historical type survived contact with an engine.

7.2 Ledgers as process data

Experimentalists already record choices. Political Swarm records *procedure*: Speaker rulings, Opposition objections with named remedies, divisions, minority reports, reconnaissance findings, returned assignments, mutual-aid packets, bids, awards, escrow releases, refused rescues. Those traces are closer to legislative minutes or to market microstructure than to a pass/fail. They make it possible to ask not only whether a constitution delivered, but how it failed: cycling, orphaned ownership, under-provided public goods, an assessor captured by a success fee (forbidden here by construction).

7.3 Constitutional floors as coded vetoes

A large part of constitutional theory is about what majorities must not do (Buchanan and Tullock, 1962; Dicey, 1885; Madison, 1788). Encoding that as a non-waivable finding set, checked before merge, is a precise analogue of a hard judicial or constitutional constraint. It is also a warning: the ground patterns are broad, and a false match ("auth" inside "author") can only be cleared with counter-evidence. Constitutional text has the same problem. The laboratory makes the false-positive visible.

7.4 Mixed government, knowledge, and recall

The three doctrines reconstruct three classic answers to the coordination problem. Westminster mixed government separates proposal, execution, scrutiny, and adjudication (Bagehot, 1867; Montesquieu, 1748). Council socialism tries to keep coordination recallable so that it does not become a class (Marx, 1871; Pannekoek, 1950). Markets treat knowledge as dispersed and use prices and contract rather than a planner (Coase, 1937; Hayek, 1945). Running them on one engine does not adjudicate those traditions. It shows that each answer can be made into a protocol with characteristic failure modes. That is closer to what Epstein (1999) asked of generative models—can you grow the phenomenon?—than to a horse race.

7.5 What the method cannot claim

LLM agents do not have interests, constituencies, or legitimacy. There is no preference aggregation theorem here, and no welfare comparison. External validity to human legislatures, firms, or movements is not on offer. The analogy is institutional, not ontological: these are *as-if* parliaments, councils, and markets. Treating a cheap model failing a nuclear-data parser as evidence about socialism would be a category error the laboratory is designed to make obvious.

Analogical inference, used carefully, is still inference. If exclusive file ownership plus an unreformed veto produces orphaned work, one has a mechanism that *could* operate wherever property in agenda items is exclusive and succession on abandonment is undefined. If a treasury levies for public goods and never spends, one has a mechanism that could operate wherever earmarks exist without a disbursement rule. Those are statements about coded institutions, offered to theorists of human institutions as hypotheses, not as results about humans. Greif (2006) studies historical institutions with a similar division of labour: the model is not the city; it is a claim about a mechanism the city might have used.

A second boundary is welfare. Nothing in the oracle is a social-welfare function. Managerial 11/11 is not "hierarchy is just." Democratic incompleteness is not "gridlock is illegitimate." The only normative

commitment in the runtime is the set of non-waivable grounds, and those are safety engineering, not political philosophy, even when they are written in constitutional vocabulary.

8 Limitations and ethics

The empirical limitations are severe and should govern citation. One sweep is one draw from a nondeterministic process. The reported sweep was not fully prompt-neutral. Several essay institutions are unwired (Appendix A). Doctrines differ in institutional call volume by design, which confounds "governance" with "more reviewers" unless process evidence is read alongside scores. OpenRouter availability and pricing are external. Verification categories are repository-configured.

The ethical limitations are structural. Worktrees isolate Git history; they do not sandbox the network or the filesystem. Apply protection is procedural. Humans remain mandatory reviewers of untrusted code. Benchmark spend is real money. The live dashboard can display historical propaganda images; they are not used in this paper and are not part of the identification strategy.

A subtler ethical issue is anthropomorphism. Naming a function `Speaker` does not create a legislature. The prose in this paper uses institutional names because the code does, and then immediately qualifies them as stylized types. Readers who skip the qualifiers will over-read. That is a communication risk the project accepts by choosing political vocabulary at all.

There is also a research-ethics point about evaluation. Hidden oracles and blinded scoring are good design. They are not licences to treat model labour as costless or to hide the dollar cost of a sweep. The table reports spend because spend is part of the institutional tax. A doctrine that "wins" by exhausting a budget another doctrine did not get was not compared. The 22 September fairness fixes exist because that comparison had not been clean.

9 Conclusion

Political science has better theories of institutions than it has laboratories in which those theories can be installed as alternative operating systems for the same task. Political Swarm is one such laboratory: three political doctrines and two controls, a shared engine, non-waivable floors, a hidden oracle, and a ledger that records procedure rather than only product. The 21 September 2026 sweep shows that the laboratory functions and that institutional bugs are scientifically informative. It does not rank regimes.

The replication path is the repository. A tagged engine, an identical starting task, a fixed roster and budget, sequential execution, the eleven-check oracle, and the two-pass rubric are enough to turn this methods paper into a results paper—provided the neutrality tests stay green and N is no longer one. Until then, the honest claim is narrower, and sufficient: institutions can be written as code, and when they are, they can be compared.

That claim is addressed to two audiences who do not usually share a methods section. Computational social scientists already run agents; they have not usually run constitutions. Comparativists already theorise constitutions; they have not usually been able to install them. If the laboratory is useful, it will be because those two literatures can argue, in the same table, about whether a Speaker who cannot strike the last motion, a council that may return work, and a market that cannot waive a safety floor are different

enough to matter. The argument will have to be about ledgers, not about which political vocabulary one finds congenial.

A Implemented versus essay-only institutions

Claims in the main text about "how the institution works" are restricted to mechanisms that the orchestrator contract and strategy modules actually run. The design essays are statements of intent. Table 4 lists the gaps decided as disclosed-not-built on 22 September 2026, plus related known limitations.

Table 4: Essay institutions that do not run, or do not run as the essay describes.

Doctrine	Gap
Democratic	Two-chamber vote: `constitutional_amendment` computes two-thirds of the Commons as advice and always goes to a human. The House of Review never votes on a motion.
Democratic	No Security Committee. `security_boundary` is advice plus a high-severity Opposition objection, then a human.
Democratic	Commons division is a formula without floor deliberation. Prime minister, Question Time, and alternative programme are nominal.
Socialist	Recall grounds beyond misallocation are not enforced. Lost confidence, concealed evidence, and council recall of the coordinator are unimplemented.
Socialist	The deciding coordinator is one persona. Planner recall takes effect on the next run.
Capitalist	Public-goods and venture funds accumulate levies but issue no grants.
Capitalist	Credit is never repaid. Warranty periods are not counted down; chargeback is not enforced. Retention may be forfeited although it was never credited.
Capitalist	Bid-collusion detection is informational. Arbitration rarely changes outcomes.
Controls	Managerial and single-agent risk class is recorded, not acted on.

Mechanisms that *are* wired, and that therefore license the main-text narrative, include: democratic office constitution from the run identifier; Speaker screening and mandate amendment; Opposition review with a required remedy; formulaic Commons votes; House of Review on findings; minority-report restoration; socialist reconnaissance, subsidiarity routing, assignment return, mutual aid, dissent preservation, and the commons report; capitalist bidding and award scoring, capital accounts, contractual assessment, antitrust self-dealing blocks, and the shared non-waivable floor.

B Oracle checks and scoring

The evaluator-owned oracle (`benchmark/oracle/checks.py`) runs eleven checks:

1. Pure-Fe reproduction against the elemental curve.
2. Fifty-fifty atomic Fe–Ni mixture equals the arithmetic mean.

3. Weight-percent to atomic-fraction conversion.
4. Composition normalisation.
5. Unsupported element flagged, not silently filled.
6. Interpolation accuracy on a synthetic grid.
7. No extrapolation beyond native data range.
8. CSV schema and values.
9. Plot generation (PNG above a minimum size).
10. Logarithmic neutron-energy axis.
11. Provenance reachable from program output.

Pass 1 of the scoring rubric weights numerical correctness 30 percent, nuclear-data handling 20 percent, verification 15 percent, documentation 10 percent, simplicity 10 percent, plots 5 percent, provenance 5 percent, and resource efficiency 5 percent, and is conducted on anonymized candidate directories. Pass 2 joins strategy names and ledger process evidence without revising Pass 1.

C Replication

The laboratory is the public repository `https://github.com/dgcarr/political-swarm` (Apache License 2.0). From `mcp-server/`, with an OpenRouter key configured:

```
npm run benchmark -- --label controlled-run \
  --budget-usd 8 --max-wall-clock-min 150
```

This invokes real models and costs money. Use identical configuration, task commit, budget, environment, and scoring for every strategy. Resume an interrupted sweep with `-resume`. Sequential execution is the comparison mode; `-concurrent` is stress only. Compile this paper with `pdflatex`, `bibtex`, and two further `pdflatex` passes in `papers/institutions-as-code/`.

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